

11 (**short**) Figure 8 shows a schematic diagram of a particle of mass m tethered to fixed supports by springs of stiffness k and $8k$ as shown, allowing for motion only in the x - y plane.

(a) Determine by inspection the normal modes of the system and obtain expressions for the natural frequencies in terms of the system parameters. [4]

(b) The system is set in motion with the initial conditions $x = 0$, $y = 0$, $\dot{x} = 0$, $\dot{y} = 1$ at time $t = 0$. Derive an expression for the subsequent motion of the particle as a function of time. [6]

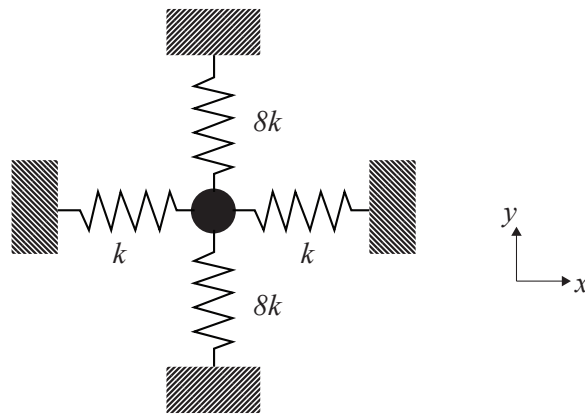


Fig. 8